

FLAGSHIP SOLUTION SCALING TRACK PROFILE

Barley-Enriched Balady Bread for a Climate-Resilient Egypt

Scaling for Impact Science Program | Area of Work 2

Grand Challenge 3 · Drive Demand, Shape Behavior & Activate Policy for Nutrition

Flagship 3.1 · Catalyzing Demand for Nutritious Diets through Markets, Institutions & Policy

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Foreword

CGIAR SCALING FOR IMPACT · AREA OF WORK 2 — PATHWAYS TO SCALE IN AGRIFOOD SYSTEMS

CGIAR produces far more scale-ready innovation than currently reaches farmers at scale. Closing that gap is the purpose of the CGIAR Scaling for Impact (S4I) Science Program. Within S4I, Area of Work 2 — Pathways to Scale in Agrifood Systems — is organized around three Grand Challenges that mark where scaling can unlock the largest development returns across food, land, and water systems in a climate crisis. These Grand Challenges were defined through an evidence-led process — drawing on the CGIAR Impact Report 2022–2024, Initiative reporting, and an analysis of eleven major donor strategies — so that each rests on demonstrated scaling potential and connects directly to the priorities of CGIAR's funders.

Under each Grand Challenge sits a focused set of **Scaling Flagships** that "dock with" and support CGIAR's Science Programs — taking mature innovation bundles and accelerating their delivery and scaling systems:

- **Grand Challenge 1 — Close the Climate Adaptation Gap** through scalable climate services and smart water management

Flagship 1.1 — Digital Climate Services for Resilient Farm & Enterprise Management

Flagship 1.2 — Scaling Smart Irrigation & Climate-Resilient Water Systems

- **Grand Challenge 2 — Deliver Inputs and Services at Scale** through inclusive market systems

Flagship 2.1 — Scaling Resilient, Nutritious & Biodiverse Seed & Soil Systems

Flagship 2.2 — Entrepreneurship through Mechanization & Postharvest Services

Flagship 2.3 — Genetics, Inputs & Services for Animals & Aquaculture

- **Grand Challenge 3 — Drive Demand, Shape Behavior & Activate Policy** for nutrition

Flagship 3.1 — Catalyzing Demand for Nutritious Diets through Markets, Institutions & Policy

Each Flagship is delivered through **Scaling Solution Tracks** — the profile that follows is one of them. Tracks are run deliberately as large, time-bound scaling campaigns, not as research projects or pilots. Experience across the portfolio shows that the binding constraint to scaling is rarely the innovation itself; it is *system integration* — the delivery and business models, finance and risk-sharing, institutional and policy alignment, and coordination among actors required to move a proven solution into widespread use. Each track therefore bundles a validated innovation with a clear delivery pathway, committed partners, and a defined geography, and is driven through and with scaling partners rather than by CGIAR alone — leveraging complementary investments to amplify reach.

This campaign approach lets S4I concentrate resources, operate in performance-based mode against measurable KPIs — farmers adopting innovations, jobs created, consumers gaining access to healthier diets, and scaling finance and supportive policy mobilized — and hold itself accountable to the S4I 2030 outcome targets and CGIAR's Impact Area goals. Across every track, scaling is pursued responsibly and inclusively, with women, youth, and marginalized groups systematically included by design.

1. Overview

Egypt faces a critical food security challenge: its national staple, *eish baladi* (balady bread), is heavily subsidized and relies on wheat imports for ~60% of its demand. This dependency costs the government over EGP 47 billion (~USD 1 billion) annually and exposes the country to the volatility of global markets. The "Barley-Enriched Balady Bread" scaling track aims to address this by integrating climate-resilient, nutritionally enhanced barley flour into Egypt's subsidized bread system – the world's largest, serving over 70 million citizens through a state-operated smart card distribution mechanism. The primary goal is to reduce wheat import dependency, improve subsidy efficiency, and create new market opportunities for farmers on marginal and saline lands, all while maintaining consumer acceptance.

The solution is a bundled innovation package operating within Egypt's existing public sector food system. It combines improved, hull-less barley varieties with new standards and protocols for industrial milling and baking, and a targeted policy shift to include barley in the national wheat subsidy program. The scaling pathway is anchored in Egypt's national balady bread subsidy system.

This scaling effort is supported by successful pilot studies conducted by ICARDA and Egypt's Food Technology Research Institute (FTRI), part of the Agricultural Research Center (ARC), which confirmed consumer acceptance of 90:10 and 80:20 wheat and barley flour blends and increased loaf output per ton of flour, creating immediate cost-efficiency gains. Core implementing partners include ARC, the Central Authority for Seed Production (CASP), and other Ministries and partners.

2. Innovation Bundle & Scaling Ambition

2.1 What is Being Scaled

- **Climate-Resilient Varieties:** Large-scale distribution of improved, hull-less barley seed varieties for cultivation on marginal and saline lands.
- **Regulatory Endorsement of Blended Flour:** A new standard for wheat-barley flour blends is currently undergoing the official endorsement process.
- **Enabling Policy Framework:** Engagement on providing farmers with a guaranteed procurement price for barley to incentivize production for the balady bread market, and on the policy and regulatory steps required for a system-wide transformation.
- **Consumer Awareness Campaign:** A national communications campaign to promote the nutritional and national food security benefits of barley-enriched bread.

2.2 Evidence of Impact

- **Consumer Acceptance:** Sensory tests show that blends of 10:90 and 20:80 (barley:wheat) are accepted by consumers in terms of taste, aroma, and texture, and there has been an increased demand for barley-fortified bread.
- **Cost-Effectiveness:** Barley flour increases water absorption during dough preparation, resulting in a higher loaf output per ton of flour, reducing the cost per loaf compared to wheat-only flour.
- **Nutritional Enhancement:** Lab tests indicate that wheat-barley bread has increased protein and fiber content compared to traditional wheat bread.

2.3 Scaling Targets

Outcome Metric	2026 Target	Long-term Goal (3–5 years)
Consumers reached	1million consumers benefiting from awareness campaigns and demonstrations of the innovation.	~5 million consumers (~10% of subsidized bread beneficiaries) with improved access to nutritious food options.
Area under improved barley (ha)	Establishment of foundation seed production and 20 demonstration fields (~10 ha).	50,000 - 70,000 hectares of climate-resilient barley cultivated on reclaimed/marginal lands.
Wheat import reduction	Baseline established and data collection on potential savings.	1–2 million tons of wheat saved annually, reducing import expenditures by hundreds of millions of USD.
Jobs supported	Number of farmers engaged in barley farming.	New agrifood sector jobs created across the barley production, milling baking, and distribution value chain.
Investment leveraged	CGIAR Sustainable Farming and mapped bilaterals	Alignment and co-investment with national land reclamation (~1.5 million feddans) and subsidy programs.

3. Scaling Pathway & Theory of Change

3.1 Pathway Strategy The core scaling strategy works within Egypt's existing state-controlled subsidy system, a fifty-year-old national institution representing approximately USD 2.1 billion in annual government expenditure. In this system, MoSIT mandates flour specifications for all subsidized mills and bakeries, GASC manages grain procurement at a government-set price, and bread is distributed to citizens at a fixed price through the national smart card system. Mills and bakeries are fee-for-service processors operating under ministerial mandate, not autonomous commercial actors. This architecture means that once the regulatory and supply conditions are met, a single policy decision triggers system-wide implementation across the entire existing infrastructure.

The key steps are:

- **Secure Regulatory Endorsement:** Support the standards endorsement process for wheat-barley flour blends and wheat-barley balady bread.
- **Create Supply:** Scale barley production on marginal lands through certified seed distribution, with the farmer incentivization model mirroring the existing wheat procurement system where the General Authority for Supply Commodities (GASC) provides a guaranteed fixed purchase price that gives farmers certainty.
- **Enable Mill and Bakery Readiness:** Work with multiple partners to develop and formalize quality assurance protocols and standard operating procedures for milling, blending, and baking wheat-barley balady bread.
- **Generate Demand:** Launch a national communications campaign to create consumer awareness of barley as a healthy staple, thereby encouraging consumer acceptance.

3.2 Theory of Change If the EOS standard for wheat-barley flour blends is endorsed, providing the regulatory foundation for system-wide adoption, **and if** a reliable supply chain of climate-resilient barley is established through seed multiplication and contract farming on marginal lands, with GASC providing a guaranteed

procurement price, **then** mills and bakeries operating under MoSIT mandate will adopt standardized blending and production processes, **leading to** the availability of affordable, nutritious barley-enriched bread through the existing smart card distribution system, **which will ultimately contribute** to reduced wheat import dependency, improved public health outcomes, enhanced resilience of farming livelihoods in marginal areas, and greater efficiency of public subsidy expenditure.

3.3 Key Activities Funded by S4I

- **Regulatory & Institutional support:** Supporting the standards endorsement process; generating technical evidence base for government policymakers; developing a clear policy roadmap for subsidy integration.
- **Socio-Economic & MELIA:** Establishing MELIA baselines; developing a framework of granular institutional change milestones; conducting farmer livelihood and incentive analysis; expand consumer acceptance field studies.
- **Seed Systems & Production Scale-Up:** Supporting ARC and CASP's seed multiplication program; targeting barley expansion in marginal and saline areas ahead of the November 2026 planting window.
- **Communications & Demand Creation:** Implementing a consumer awareness campaign in collaboration with the Ministry's communications department to reach at least 1 million people.
- **Value Chain Coordination:** Strengthening linkages between all actors in the value chain to ensure coherent sequencing across the regulatory, production, and implementation pathways.

4. Partners & Roles

Partner Type	Specific Partners & Quantities	Role in Scaling Pathway
Demand Partners	Ministry of Supply and Internal Trade (MoSIT) (1); Health-conscious urban consumers (target of 1.5M); Farming communities (20+ demo fields).	MoSIT provides the essential policy leadership, procurement signals, and subsidy interface that mandates adoption. Farmers supply the raw material.
Scaling Partners	Ministry of Agriculture & Land Reclamation (MoALR) (1); Agricultural Research Center (ARC) & Food Technology Research Institute (FTRI) (2); CGIAR Program (Sustainable Farming) (2).	MoALR, ARC, and FTRI provide technical backstopping, seed multiplication, quality standards, and extension. CGIAR programs provide complementary funding, research, and scaling platforms.
Funding Partners	CGIAR S4I Program; CGIAR Sustainable Farming Program; ICARDA-ARC Bilateral Project.	Provide catalytic and complementary resources to de-risk the pathway and enable the system-level changes required for scaling.

Funding Model: S4I catalytic funding covers the essential enabling activities, regulatory support, policy engagement, socio-economic research, communications, and multi-stakeholder coordination, while the Egyptian government's existing subsidy infrastructure, and complementary donor projects, provide the financial foundation for production and distribution at scale.

5. Enabling Environment & Risks

5.1 Opportunities

- **Strong Policy Alignment:** The track is explicitly aligned with Egypt Vision 2030, the National Climate Strategy 2050, and the Sustainable Agriculture Development Strategy 2030, creating a powerful enabling environment.
- **Existing Infrastructure:** The world's largest subsidized bread system, with centralized flour mills and a network of bakeries, provides a ready-made, large-scale delivery channel.
- **Government Alignment:** The Government has a clear, stated interest in reducing the fiscal burden of wheat subsidies and has previously explored alternative blends, signaling strong government demand for a solution.
- **Complementary Investments:** The initiative can leverage significant ongoing public and donor investments in land reclamation (~1.5 million feddans), irrigation, and the national wheat-fortification program.

5.2 Challenges & Mitigation

- **Political/Regulatory Risk:** Altering a national staple involves slow policy cycles and potential political sensitivity.
- *Mitigation:* The project works within the existing system architecture and does not require changes to consumer pricing, daily bread quotas, or the equity of access through the smart card system, barley integration strengthens rather than threatens the subsidy system's fiscal sustainability.
- **Regulatory Timeline:** The standards endorsement process may extend beyond 2026.
- *Mitigation:* The project has designed its workstreams so that consumer acceptance studies, seed multiplication, and farmer engagement can proceed in parallel, ensuring that no time is lost while the regulatory pathway is conducted.
- **Seed Supply and Production Scale-Up:** Building a domestic barley supply at the volumes required for national scale takes multiple growing seasons.
- *Mitigation:* scaling certified seed production while also investigating near-term sourcing of existing uncontracted domestic barley or procured international supply.

6. Learning & Adaptation

6.1 Monitoring Focus

- **Process outcomes:** Effectiveness of policy engagement with MoSIT; speed of standards endorsement and operationalisation; functionality of the contract farming model.
- **Behavior change:** Adoption of barley cultivation by farmers in target areas; willingness of consumers to accept the new bread.
- **System outcomes:** Changes in subsidy cost per loaf; volume of wheat imports displaced; increases in barley acreage in marginal lands.
- **Attribution:** The MELIA framework will be designed to demonstrate how changes are linked to the S4I intervention, tracking the 1 million consumers reached through awareness campaigns and pilot tastings.

6.2 Adaptive Management

- **Quarterly Check-ins:** Regular meetings with the S4I team to review progress against the work plan and MELIA data, allowing for course correction.
- **Multi-Stakeholder Reviews:** Annual learning briefs and workshops with key partners to review evidence, share lessons, and jointly adapt the scaling strategy.
- **Peer Learning:** Facilitated connections with other S4I tracks and experts (e.g., on subsidy reform) to bring in external perspectives and best practices.

7. Next Steps & Timeline (2026)

Quarter	Key Activities
Q1	Finalize detailed Theory of Change and 2026 work plan with partners. . Establish MELIA baselines.
Q2	Formalise partnership arrangements with ARC/FTRI. Expand consumer acceptance field studies to rural and barley-growing communities. Monitor demonstration fields and foundation seed production.
Q3	Certify the multiplied seed. Generate consumer acceptance evidence from expanded field studies. Conduct farmer engagement and geographic targeting for barley expansion areas. Support the preparation of technical findings for MSIT and the Chamber of Cereal Industry.
Q4	Report on institutional milestones achieved. Consolidate evidence base for regulatory and policy engagement. Plan for expansion of activities into 2027.

Communications & Knowledge: Key products will include a series of policy briefs for partner institutions, learning briefs for the CGIAR community, and mass media content (radio spots, social media content) for the national consumer awareness campaign.

8. Contact

- **Andrea Visioni (ICARDA-Morocco)** | a.visioni@cgiar.org
- **Rhiannon Crichton (ICARDA-Ireland)** | r.crichton@cgiar.org